

Carbon Capture Utilization and Storage Market Size, Share, Growth 2026-2035

The carbon capture utilization and storage market was valued at USD 5.7 billion in 2025 and is anticipated to reach USD 35.5 billion by the end of 2035, expanding at a compound annual growth rate (CAGR) of around 20.1% during the forecast period of 2026-2035. Strong governmental support for decarbonization, rising industrial investments in emission reduction technologies, and increasing deployment of carbon management infrastructure are key factors driving market expansion. Growing commitments to net-zero emissions across industries are expected to sustain long-term market growth.

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Carbon Capture Utilization and Storage Industry Demand

Carbon Capture Utilization and Storage refers to a group of technologies designed to capture carbon dioxide emissions from industrial facilities and power plants, transport the captured carbon, and either utilize it in industrial applications or permanently store it in underground geological formations. The technology plays a significant role in reducing greenhouse gas emissions from sectors that are difficult to decarbonize using renewable energy alone.

Demand for CCUS solutions is increasing due to stricter environmental regulations, corporate sustainability commitments, and the global transition toward low-carbon economies. Industries such as cement, steel, chemicals, and power generation are adopting CCUS technologies to reduce emissions while maintaining operational efficiency. Improvements in capture technologies, expanding carbon transportation networks, and supportive government incentives are further accelerating market adoption. Continuous innovation aimed at lowering operational costs and improving capture efficiency is also strengthening industry demand.

Carbon Capture Utilization and Storage Market: Growth Drivers & Key Restraint

Growth Drivers

- **Global Net-Zero Emission Commitments:** Governments and major industrial organizations are implementing ambitious climate policies to reduce carbon emissions. CCUS has become a vital solution for achieving emission reduction targets, particularly in heavy industries where complete electrification remains challenging.
- **Technological Advancements in Carbon Capture:** Continuous improvements in capture materials, membrane technologies, chemical looping processes, and digital monitoring systems have enhanced the efficiency and reliability of carbon capture

operations. These innovations are helping industries reduce operational costs while improving overall project performance.

- **Rising Investments and Government Incentives:** Public funding programs, tax credits, carbon pricing mechanisms, and private-sector investments are encouraging the development of commercial-scale CCUS projects. Strategic partnerships between governments and industries are accelerating infrastructure development and technology deployment worldwide.

Restraint

- **High Capital Investment and Infrastructure Requirements:** Despite growing adoption, large-scale CCUS implementation requires significant investment in capture facilities, transportation pipelines, compression systems, and long-term storage infrastructure. These high upfront costs remain a major challenge, particularly for small and medium-sized industrial operators.

Carbon Capture Utilization and Storage Market: Segment Analysis

Segment Analysis by Service

Capture represents the largest service segment as it forms the foundation of every CCUS project by separating carbon dioxide from industrial emission sources. Solvents and Sorbents continue to experience strong demand because of their effectiveness in post-combustion capture applications. Chemical Looping is attracting increasing attention due to its ability to improve capture efficiency while reducing energy consumption. Membranes are gaining popularity because of their compact design and operational simplicity in selected industrial processes. Transportation services are expanding alongside the development of pipeline and shipping infrastructure required to move captured carbon safely. Utilization is witnessing growing demand as industries convert captured carbon into fuels, chemicals, construction materials, and other value-added products. Storage remains a critical component of the market, supporting permanent geological sequestration to achieve long-term emission reduction goals.

Segment Analysis by Technology

Solvents and Sorbents continue to dominate technology adoption owing to their commercial maturity and broad industrial acceptance. Chemical Looping is emerging as an efficient alternative that minimizes energy penalties during capture operations. Membrane Technologies are steadily expanding due to ongoing improvements in material performance and process efficiency. The Others category, including cryogenic separation and advanced adsorption technologies, is gaining momentum as research institutions and technology developers introduce next-generation carbon capture solutions.

Segment Analysis by End-User Industry

The Oil and Gas industry remains one of the largest adopters of CCUS technologies, utilizing captured carbon for enhanced oil recovery and emission management. Power Generation companies continue integrating carbon capture systems to reduce emissions from fossil fuel-based electricity production. The Chemical and Petrochemical industry increasingly adopts CCUS to improve sustainability while meeting regulatory requirements. Cement manufacturers are implementing carbon capture technologies to address process-related emissions that are difficult to eliminate through conventional methods. The Iron and Steel sector is expanding investment in CCUS to support cleaner production practices. Other industrial sectors are also adopting these technologies as sustainability becomes an important operational priority.

Carbon Capture Utilization and Storage Market: Regional Insights

North America

North America leads the carbon capture utilization and storage market due to strong government incentives, established carbon transportation infrastructure, and early commercialization of CCUS technologies. Extensive investment by energy companies, favorable regulatory frameworks, and large-scale industrial decarbonization initiatives continue to support regional market growth.

Europe

Europe maintains a significant market position through aggressive climate policies, expanding carbon pricing mechanisms, and substantial investment in industrial decarbonization projects. Increasing collaboration between governments, research institutions, and private companies is accelerating the deployment of advanced carbon capture and storage facilities across the region.

Asia-Pacific (APAC)

Asia-Pacific is emerging as one of the fastest-growing markets as rapidly industrializing economies focus on reducing emissions while supporting economic development. Rising investments in clean energy infrastructure, expanding manufacturing activities, government-backed carbon reduction programs, and increasing participation from heavy industries are creating strong growth opportunities throughout the region.

Top Players in the Carbon Capture Utilization and Storage Market

Major companies operating in the carbon capture utilization and storage market include Shell plc (UK), Exxon Mobil Corporation (U.S.), Equinor ASA (Norway), TotalEnergies SE (France), Fluor Corporation (U.S.), Linde plc (UK), Mitsubishi Heavy Industries, Ltd. (Japan), JGC Holdings Corporation (Japan), SLB (U.S.), Aker Solutions ASA (Norway), Global Cement and Concrete Association (UK), Global CCS Institute (Australia), CarbonFree (U.S.), and Univar Solutions (U.S.). These organizations are actively expanding their CCUS capabilities through strategic partnerships, technology innovation, pilot projects, infrastructure

development, and investments in large-scale carbon management solutions. Their focus on improving capture efficiency, transportation networks, utilization pathways, and permanent storage capacity continues to strengthen the commercial adoption of CCUS technologies across multiple industrial sectors.

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